

Functional Constraints

Flame competition constraint within pocket regions.

Penalty coefficient 0.0100

Maximum allowable heat flux density ratio (max/min) 1000.00

The linear burning rate of the first conditional fuel is greater than that of the second.

Penalty coefficient 0.0100

Constraint on maximum kinetic flame heat flux density.

Penalty coefficient 0.0100

Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09

Maximum kinetic flame heat flux density in skeleton structure 1.00e+08

Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08

Constraint on skeleton thickness to pore diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to pore diameter ratio 3.00

Constraint on skeleton thickness to large oxidizer particle diameter ratio.

Penalty coefficient 0.0100

Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range

Minimum surface temperature 600 K

Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 456

Mutation force: 0.5

Crossover probability: 0.9

Number of threads: 24

Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Shared Parameters (All Compositions)

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 211675626577.5605

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas phase reactions in inter-pocket medium 249990.95115155267

Parameter bounds [5.000E+004; 2.500E+005]

Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 305838.21810507396

Parameter bounds [1.000E+005; 1.000E+013]

Activation energy for gas-phase reactions in pocket region outside skeleton 50000.02118900488

Parameter bounds [5.000E+004; 2.500E+005]

Reaction order for gas-phase chemistry in inter-pocket medium 2.499999186004348

Parameter bounds [0.000E+000; 2.500E+000]

Reaction order for gas-phase chemistry outside skeleton structure 0.9256963149914357

Parameter bounds [0.000E+000; 2.500E+000]

Enthalpy difference between vaporization and binder decomposition 448054.5874015256

Parameter bounds [-1.000E+007; 1.000E+007]

Diffusion flame height correlation coefficient 2.0755

Parameter bounds [1.000E-003; 1.000E+001]

Power law exponent for burning rate dependency on A (skeleton thickness) 0.6136

Parameter bounds [0.000E+000; 3.000E+000]

Power law exponent for burning rate dependency on B (pore diameter) 0.6125

Parameter bounds [0.000E+000; 3.000E+000]

Radiation temperature correlation coefficient (average layer temperature) 0.0000

Parameter bounds [0.000E+000; 1.000E+000]

Diffusion-flame standoff pressure-factor C_{rxn} 0.00158182

Parameter bounds [0.000E+000; 5.000E+001]

Diffusion-flame standoff size-exponent m 0.396517

Parameter bounds [0.000E+000; 2.000E+000]

Bimodal-packing heat-feedback K_{pack} 9.74589

Parameter bounds [0.000E+000; 1.000E+001]
Diffusion-flame standoff pressure-exponent n_p 3.99999
Parameter bounds [2.000E+000; 4.000E+000]
WSB condensed-phase reaction K_wsb 0.166477
Parameter bounds [0.000E+000; 5.000E+000]
AP premixed-flame conductance K_AP 4253.28
Parameter bounds [0.000E+000; 5.000E+003]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9995343428412.662
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 184381.9144082219
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 15960600.862602426
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 50038.747115051214
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.1497746387100975E-06
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 3.572854354008596E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 8.764631335340819E-11
Parameter bounds [1.000E-012; 1.000E-005]

Bas_1 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9966408751699.271
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 136596.15739770103
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 2493783186.2745953
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 89232.48358405803
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 1.3635039030900078
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 6.82288378980007E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.1352967666354118E-10
Parameter bounds [1.000E-012; 1.000E-005]

Bas_0 - Condensed Phase Parameters

Pre-exponential coefficient for condensed phase reactions 9995819464181.445
Parameter bounds [1.000E+000; 1.000E+013]
Activation energy for condensed phase reactions 184381.1843660333
Parameter bounds [5.000E+004; 3.000E+005]
Pre-exponential factor for gas-phase reactions within skeleton structure 180992621.68298987
Parameter bounds [1.000E+005; 1.000E+013]
Activation energy for gas-phase reactions within skeleton structure 55138.61464851505
Parameter bounds [5.000E+004; 2.500E+005]
Reaction order for gas-phase chemistry in skeleton structure 0.6079453345720688
Parameter bounds [0.000E+000; 2.500E+000]
Coefficient A for metal combustion heat flux density in skeleton structure, 3.174655416843609E-06
Parameter bounds [1.000E-010; 1.000E-003]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.0626242584887369E-06
Parameter bounds [1.000E-012; 1.000E-005]

Optimization Results

Objective function value 0.0144

Total penalty from constraint violations 0.0000

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s
Calculated burning rate 15.03 mm/s

First Conditional Fuel

Surface temperature 820.96 K
Linear burning rate 11.19 mm/s
Mass decomposition rate $18.56 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.29\text{e}+07 \text{ W/m}^2$
Kinetic flame height 12.62 μm
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,263.98 K
Binder sublimation heat flux density $2.29\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0011 W/m^2

Secondary Conditional Fuel

Surface temperature 810.98 K
Linear burning rate 8.02 mm/s
Mass decomposition rate $13.31 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $188,989.63 \text{ W/m}^2$
Kinetic flame height 822.28 μm
Average kinetic flame density 2.5 kg/m^3
Average kinetic flame temperature 1,587.99 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.77\text{e}+06 \text{ W/m}^2$
Kinetic flame height 8.89 μm
Average kinetic flame density 3.3 kg/m^3
Average kinetic flame temperature 1,200.99 K
Heat flux density due to metal combustion, $3.48\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.44 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.033 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
Skeleton layer thickness $6.13\text{e}-05 \text{ m}$
Pore diameter $2.04\text{e}-05 \text{ m}$
Diffusion flame heat flux density $62,778.15 \text{ W/m}^2$
Diffusion flame height 4,449.04 μm
Heat flux density to surface within skeletal structure $4.4\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $121,155.11 \text{ W/m}^2$
Total heat flux density to surface $1.47\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $1.47\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0031 W/m^2

At pressure 4.06 MPa

Experimental burning rate 35.09 mm/s
Calculated burning rate 36.17 mm/s

First Conditional Fuel

Surface temperature 878.65 K
Linear burning rate 65.92 mm/s
Mass decomposition rate $109.36 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.44\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.96 μm
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,292.83 K
Binder sublimation heat flux density $1.44\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.003 W/m^2

Secondary Conditional Fuel

Surface temperature 832.63 K

Linear burning rate 16.34 mm/s

Mass decomposition rate $27.1 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 340,975.74 W/m²

Kinetic flame height 449.41 μm

Average kinetic flame density 10.07 kg/m³

Average kinetic flame temperature 1,598.82 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.03\text{e}+07 \text{ W/m}^2$

Kinetic flame height 7.4 μm

Average kinetic flame density 13.28 kg/m³

Average kinetic flame temperature 1,211.82 K

Heat flux density due to metal combustion, $5.01\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity 0.42 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 0.021 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness $3.96\text{e}-05 \text{ m}$

Pore diameter $1.32\text{e}-05 \text{ m}$

Diffusion flame heat flux density 122,516.5 W/m²

Diffusion flame height 2,262.04 μm

Heat flux density to surface within skeletal structure $5.04\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure 228,335.67 W/m²

Total heat flux density to surface $1.8\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $1.8\text{e}+07 \text{ W/m}^2$

Surface heat balance error 0.0037 W/m²

At pressure 6.5 MPa

Experimental burning rate 44.85 mm/s

Calculated burning rate 44.15 mm/s

First Conditional Fuel

Surface temperature 899.99 K

Linear burning rate 119.93 mm/s

Mass decomposition rate $198.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.69\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.04 μm

Average kinetic flame density 11.2 kg/m³

Average kinetic flame temperature 2,303.5 K

Binder sublimation heat flux density $2.69\text{e}+08 \text{ W/m}^2$

Surface heat balance error 0.021 W/m²

Secondary Conditional Fuel

Surface temperature 837.83 K

Linear burning rate 19.27 mm/s

Mass decomposition rate $31.97 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 447,892.88 W/m²

Kinetic flame height 340.97 μm

Average kinetic flame density 16.11 kg/m³

Average kinetic flame temperature 1,601.41 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.16\text{e}+07 \text{ W/m}^2$

Kinetic flame height 6.48 μm

Average kinetic flame density 21.24 kg/m³

Average kinetic flame temperature 1,214.41 K

Heat flux density due to metal combustion, $5.46\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity 0.42 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 0.019 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.58e-05 m
Pore diameter 1.19e-05 m
Diffusion flame heat flux density 104,686.86 W/m²
Diffusion flame height 2,642.33 μm
Heat flux density to surface within skeletal structure 4.64e+06 W/m²
Heat flux density to surface outside skeletal structure 326,209.46 W/m²
Total heat flux density to surface 1.75e+07 W/m²
Binder sublimation heat flux density 1.75e+07 W/m²
Surface heat balance error -0.0032 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s
Calculated burning rate 14.93 mm/s

First Conditional Fuel

Surface temperature 820.96 K
Linear burning rate 11.19 mm/s
Mass decomposition rate 18.56 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 2.29e+07 W/m²
Kinetic flame height 12.62 μm
Average kinetic flame density 1.75 kg/m³
Average kinetic flame temperature 2,263.98 K
Binder sublimation heat flux density 2.29e+07 W/m²
Surface heat balance error -0.0011 W/m²

Secondary Conditional Fuel

Surface temperature 820.96 K
Linear burning rate 11.19 mm/s
Mass decomposition rate 18.56 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 135,882.41 W/m²
Kinetic flame height 1,136.31 μm
Average kinetic flame density 2.49 kg/m³
Average kinetic flame temperature 1,592.98 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.21e+07 W/m²
Kinetic flame height 2.9 μm
Average kinetic flame density 2.3 kg/m³
Average kinetic flame temperature 1,721.98 K
Heat flux density due to metal combustion, 3.7e+06 W/m²
Effective thermal conductivity 0.39 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.16 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 5e-05 m
Pore diameter 1.67e-05 m
Diffusion flame heat flux density 119,989.1 W/m²
Diffusion flame height 2,377.75 μm
Heat flux density to surface within skeletal structure 2.06e+07 W/m²
Heat flux density to surface outside skeletal structure 93,363.38 W/m²
Total heat flux density to surface 2.08e+07 W/m²
Binder sublimation heat flux density 2.08e+07 W/m²
Surface heat balance error 0.0015 W/m²

At pressure 4.06 MPa

Experimental burning rate 37.34 mm/s
Calculated burning rate 38.31 mm/s

First Conditional Fuel

Surface temperature 878.65 K
Linear burning rate 65.92 mm/s
Mass decomposition rate $109.36 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.44\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.96 \mu\text{m}$
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,292.83 K
Binder sublimation heat flux density $1.44\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.003 W/m^2

Secondary Conditional Fuel

Surface temperature 839.2 K
Linear burning rate 20.12 mm/s
Mass decomposition rate $33.38 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $277,268.43 \text{ W/m}^2$
Kinetic flame height $550.3 \mu\text{m}$
Average kinetic flame density 10.05 kg/m^3
Average kinetic flame temperature 1,602.1 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $8.14\text{e}+07 \text{ W/m}^2$
Kinetic flame height $2.19 \mu\text{m}$
Average kinetic flame density 9.3 kg/m^3
Average kinetic flame temperature 1,731.1 K
Heat flux density due to metal combustion, $4.45\text{e}+06 \text{ W/m}^2$
Effective thermal conductivity $0.34 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $0.22 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.12 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $-5.5152085010057306\text{E}-08$
Skeleton layer thickness $3.49\text{e}-05 \text{ m}$
Pore diameter $1.16\text{e}-05 \text{ m}$

Diffusion flame heat flux density $254,773.41 \text{ W/m}^2$
Diffusion flame height $1,112.68 \mu\text{m}$
Heat flux density to surface within skeletal structure $2\text{e}+07 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $212,614.91 \text{ W/m}^2$
Total heat flux density to surface $2.22\text{e}+07 \text{ W/m}^2$
Binder sublimation heat flux density $2.22\text{e}+07 \text{ W/m}^2$
Surface heat balance error -0.0027 W/m^2

At pressure 6.5 MPa

Experimental burning rate 50.5 mm/s
Calculated burning rate 44.96 mm/s

First Conditional Fuel

Surface temperature 899.99 K
Linear burning rate 119.93 mm/s
Mass decomposition rate $198.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.69\text{e}+08 \text{ W/m}^2$
Kinetic flame height $1.04 \mu\text{m}$
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density $2.69\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.021 W/m^2

Secondary Conditional Fuel

Surface temperature 841.8 K
Linear burning rate 21.83 mm/s
Mass decomposition rate $36.22 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $395,670.75 \text{ W/m}^2$
Kinetic flame height $384.97 \mu\text{m}$

Average kinetic flame density 16.09 kg/m³
 Average kinetic flame temperature 1,603.4 K
Skeleton Structure Region
 Kinetic flame heat flux density to surface 1e+08 W/m²
 Kinetic flame height 1.78 μm
 Average kinetic flame density 14.89 kg/m³
 Average kinetic flame temperature 1,732.4 K
 Heat flux density due to metal combustion, 4.58e+06 W/m²
 Effective thermal conductivity 0.33 W/(m·K)
 Conductive thermal conductivity 0.22 W/(m·K)
 Radiative thermal conductivity 0.11 W/(m·K)
 Conductive thermal conductivity balance error -5.5152085010057306E-08
 Skeleton layer thickness 3.32e-05 m
 Pore diameter 1.11e-05 m
 Diffusion flame heat flux density 236,700.4 W/m²
 Diffusion flame height 1,196.53 μm
 Heat flux density to surface within skeletal structure 1.58e+07 W/m²
 Heat flux density to surface outside skeletal structure 335,795.06 W/m²
 Total heat flux density to surface 1.99e+07 W/m²
 Binder sublimation heat flux density 1.99e+07 W/m²
 Surface heat balance error -0.004 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa

Experimental burning rate 9.73 mm/s
 Calculated burning rate 10.22 mm/s

First Conditional Fuel

Surface temperature 820.96 K
 Linear burning rate 11.19 mm/s
 Mass decomposition rate 18.56 kg·s⁻¹·m⁻²
 Kinetic flame heat flux density to surface 2.29e+07 W/m²
 Kinetic flame height 12.62 μm
 Average kinetic flame density 1.75 kg/m³
 Average kinetic flame temperature 2,263.98 K
 Binder sublimation heat flux density 2.29e+07 W/m²
 Surface heat balance error -0.0011 W/m²

Secondary Conditional Fuel

Surface temperature 787.32 K
 Linear burning rate 3.53 mm/s
 Mass decomposition rate 5.85 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 7,226.83 W/m²
 Kinetic flame height 3,524.15 μm
 Average kinetic flame density 4.34 kg/m³
 Average kinetic flame temperature 914.66 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.65e+06 W/m²
 Kinetic flame height 9.75 μm
 Average kinetic flame density 3.91 kg/m³
 Average kinetic flame temperature 1,014.16 K
 Heat flux density due to metal combustion, 7.23e+06 W/m²
 Effective thermal conductivity 1.43 W/(m·K)
 Conductive thermal conductivity 1.41 W/(m·K)
 Radiative thermal conductivity 0.021 W/(m·K)
 Conductive thermal conductivity balance error 9.240271880983641E-10
 Skeleton layer thickness 1.02e-04 m
 Pore diameter 3.38e-05 m

Diffusion flame heat flux density 19,270.69 W/m²

Diffusion flame height 14,491.87 μm
Heat flux density to surface within skeletal structure $6.28\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $3,405.52 \text{ W/m}^2$
Total heat flux density to surface $6.3\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $6.3\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-1.89\text{e}-04 \text{ W/m}^2$

At pressure 4.06 MPa

Experimental burning rate 25.92 mm/s
Calculated burning rate 26.05 mm/s

First Conditional Fuel

Surface temperature 878.65 K
Linear burning rate 65.92 mm/s
Mass decomposition rate $109.36 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $1.44\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.96 μm
Average kinetic flame density 7.02 kg/m^3
Average kinetic flame temperature 2,292.83 K
Binder sublimation heat flux density $1.44\text{e}+08 \text{ W/m}^2$
Surface heat balance error 0.003 W/m^2

Secondary Conditional Fuel

Surface temperature 810.47 K
Linear burning rate 7.89 mm/s
Mass decomposition rate $13.08 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $11,524.81 \text{ W/m}^2$
Kinetic flame height 2,009 μm
Average kinetic flame density 17.38 kg/m^3
Average kinetic flame temperature 926.23 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $4.94\text{e}+06 \text{ W/m}^2$
Kinetic flame height 8.71 μm
Average kinetic flame density 15.69 kg/m^3
Average kinetic flame temperature 1,025.73 K
Heat flux density due to metal combustion, $1.12\text{e}+07 \text{ W/m}^2$
Effective thermal conductivity $1.42 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity $1.41 \text{ W/(m}\cdot\text{K)}$
Radiative thermal conductivity $0.013 \text{ W/(m}\cdot\text{K)}$
Conductive thermal conductivity balance error $9.240271880983641\text{E}-10$
Skeleton layer thickness $6.2\text{e}-05 \text{ m}$
Pore diameter $2.06\text{e}-05 \text{ m}$
Diffusion flame heat flux density $48,669.76 \text{ W/m}^2$
Diffusion flame height 5,690.46 μm
Heat flux density to surface within skeletal structure $8.5\text{e}+06 \text{ W/m}^2$
Heat flux density to surface outside skeletal structure $5,474.52 \text{ W/m}^2$
Total heat flux density to surface $8.55\text{e}+06 \text{ W/m}^2$
Binder sublimation heat flux density $8.55\text{e}+06 \text{ W/m}^2$
Surface heat balance error $-9.8\text{e}-04 \text{ W/m}^2$

At pressure 6.5 MPa

Experimental burning rate 36.06 mm/s
Calculated burning rate 35.49 mm/s

First Conditional Fuel

Surface temperature 899.99 K
Linear burning rate 119.93 mm/s
Mass decomposition rate $198.96 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
Kinetic flame heat flux density to surface $2.69\text{e}+08 \text{ W/m}^2$
Kinetic flame height 1.04 μm
Average kinetic flame density 11.2 kg/m^3

Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 2.69e+08 W/m²
Surface heat balance error 0.021 W/m²

Secondary Conditional Fuel

Surface temperature 819.02 K
Linear burning rate 10.49 mm/s
Mass decomposition rate 17.41 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 13,240.17 W/m²
Kinetic flame height 1,684.1 μm
Average kinetic flame density 27.72 kg/m³
Average kinetic flame temperature 930.51 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.97e+06 W/m²
Kinetic flame height 8.49 μm
Average kinetic flame density 25.05 kg/m³
Average kinetic flame temperature 1,030.01 K
Heat flux density due to metal combustion, 1.31e+07 W/m²
Effective thermal conductivity 1.42 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.011 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 5.2e-05 m
Pore diameter 1.73e-05 m
Diffusion flame heat flux density 37,003.36 W/m²
Diffusion flame height 7,461.42 μm
Heat flux density to surface within skeletal structure 9.45e+06 W/m²
Heat flux density to surface outside skeletal structure 6,331.95 W/m²
Total heat flux density to surface 9.49e+06 W/m²
Binder sublimation heat flux density 9.49e+06 W/m²
Surface heat balance error -8.37e-04 W/m²

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s
Calculated burning rate 14.3 mm/s

First Conditional Fuel

Surface temperature 606.41 K
Linear burning rate 10.3 mm/s
Mass decomposition rate 17.09 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.56e+07 W/m²
Kinetic flame height 19.92 μm
Average kinetic flame density 1.84 kg/m³
Average kinetic flame temperature 2,156.7 K
Binder sublimation heat flux density 1.56e+07 W/m²
Surface heat balance error 0.0036 W/m²

Secondary Conditional Fuel

Surface temperature 600 K
Linear burning rate 7.71 mm/s
Mass decomposition rate 12.79 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 181,739.18 W/m²
Kinetic flame height 971.17 μm
Average kinetic flame density 2.68 kg/m³
Average kinetic flame temperature 1,482.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.22e+06 W/m²
Kinetic flame height 15.94 μm

Average kinetic flame density 3.62 kg/m^3
 Average kinetic flame temperature $1,095.5 \text{ K}$
 Heat flux density due to metal combustion, $2.09\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $3.6\text{e}-06 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
 Skeleton layer thickness $1.35\text{e}-04 \text{ m}$
 Pore diameter $2.23\text{e}-09 \text{ m}$
 Diffusion flame heat flux density $71,683.8 \text{ W/m}^2$
 Diffusion flame height $4,277.12 \text{ }\mu\text{m}$
 Heat flux density to surface within skeletal structure $3.14\text{e}+06 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $113,031.9 \text{ W/m}^2$
 Total heat flux density to surface $1.07\text{e}+07 \text{ W/m}^2$
 Binder sublimation heat flux density $1.07\text{e}+07 \text{ W/m}^2$
 Surface heat balance error 0.0022 W/m^2

At pressure 4.06 MPa

Experimental burning rate 38.01 mm/s
 Calculated burning rate 38.08 mm/s

First Conditional Fuel

Surface temperature 648.67 K
 Linear burning rate 60.17 mm/s
 Mass decomposition rate $99.83 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$
 Kinetic flame heat flux density to surface $9.72\text{e}+07 \text{ W/m}^2$
 Kinetic flame height $3.15 \text{ }\mu\text{m}$
 Average kinetic flame density 7.39 kg/m^3
 Average kinetic flame temperature $2,177.84 \text{ K}$
 Binder sublimation heat flux density $9.72\text{e}+07 \text{ W/m}^2$
 Surface heat balance error -0.024 W/m^2

Secondary Conditional Fuel

Surface temperature 618.49 K
 Linear burning rate 17.49 mm/s
 Mass decomposition rate $29.01 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $295,548.88 \text{ W/m}^2$
 Kinetic flame height $590.94 \text{ }\mu\text{m}$
 Average kinetic flame density 10.79 kg/m^3
 Average kinetic flame temperature $1,491.75 \text{ K}$

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.95\text{e}+07 \text{ W/m}^2$
 Kinetic flame height $4.99 \text{ }\mu\text{m}$
 Average kinetic flame density 14.57 kg/m^3
 Average kinetic flame temperature $1,104.75 \text{ K}$
 Heat flux density due to metal combustion, $3.37\text{e}+06 \text{ W/m}^2$
 Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$
 Radiative thermal conductivity $2.18\text{e}-06 \text{ W/(m}\cdot\text{K)}$
 Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$
 Skeleton layer thickness $8.17\text{e}-05 \text{ m}$
 Pore diameter $1.35\text{e}-09 \text{ m}$
 Diffusion flame heat flux density $125,870.05 \text{ W/m}^2$
 Diffusion flame height $2,421.15 \text{ }\mu\text{m}$
 Heat flux density to surface within skeletal structure $4.58\text{e}+06 \text{ W/m}^2$
 Heat flux density to surface outside skeletal structure $236,351.56 \text{ W/m}^2$
 Total heat flux density to surface $1.67\text{e}+07 \text{ W/m}^2$
 Binder sublimation heat flux density $1.67\text{e}+07 \text{ W/m}^2$
 Surface heat balance error 0.0022 W/m^2

At pressure 6.5 MPa

Experimental burning rate 52.88 mm/s

Calculated burning rate 52.6 mm/s

First Conditional Fuel

Surface temperature 664.29 K

Linear burning rate 109.14 mm/s

Mass decomposition rate $181.06 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $1.81\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.68 μm

Average kinetic flame density 11.8 kg/m^3

Average kinetic flame temperature 2,185.64 K

Binder sublimation heat flux density $1.81\text{e}+08 \text{ W/m}^2$

Surface heat balance error -0.055 W/m^2

Secondary Conditional Fuel

Surface temperature 625.41 K

Linear burning rate 23.46 mm/s

Mass decomposition rate $38.91 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $342,065.01 \text{ W/m}^2$

Kinetic flame height 508.56 μm

Average kinetic flame density 17.25 kg/m^3

Average kinetic flame temperature 1,495.2 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $2.82\text{e}+07 \text{ W/m}^2$

Kinetic flame height 3.43 μm

Average kinetic flame density 23.28 kg/m^3

Average kinetic flame temperature 1,108.2 K

Heat flux density due to metal combustion, $3.99\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $1.82\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error $-4.092294297874943\text{E}-08$

Skeleton layer thickness $6.82\text{e}-05 \text{ m}$

Pore diameter $1.13\text{e}-09 \text{ m}$

Diffusion flame heat flux density $94,537.47 \text{ W/m}^2$

Diffusion flame height 3,216.28 μm

Heat flux density to surface within skeletal structure $5.25\text{e}+06 \text{ W/m}^2$

Heat flux density to surface outside skeletal structure $286,217.98 \text{ W/m}^2$

Total heat flux density to surface $1.98\text{e}+07 \text{ W/m}^2$

Binder sublimation heat flux density $1.98\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0051 W/m^2

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s

Calculated burning rate 5.97 mm/s

First Conditional Fuel

Surface temperature 820.96 K

Linear burning rate 11.19 mm/s

Mass decomposition rate $18.56 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $2.29\text{e}+07 \text{ W/m}^2$

Kinetic flame height 12.62 μm

Average kinetic flame density 1.75 kg/m^3

Average kinetic flame temperature 2,263.98 K

Binder sublimation heat flux density $2.29\text{e}+07 \text{ W/m}^2$

Surface heat balance error -0.0054 W/m^2

Secondary Conditional Fuel

Surface temperature 779.79 K

Linear burning rate 2.69 mm/s

Mass decomposition rate $4.46 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 559,449.24 W/m²

Kinetic flame height 283.35 μm

Average kinetic flame density 2.52 kg/m^3

Average kinetic flame temperature 1,572.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface $1.81\text{e}+06 \text{ W/m}^2$

Kinetic flame height 44.77 μm

Average kinetic flame density 3.35 kg/m^3

Average kinetic flame temperature 1,185.39 K

Heat flux density due to metal combustion, $1.56\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $5.29\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness $1.35\text{e}-04 \text{ m}$

Pore diameter $3.29\text{e}-09 \text{ m}$

Diffusion flame heat flux density 193,686 W/m²

Diffusion flame height 1,490.15 μm

Heat flux density to surface within skeletal structure 870,743.38 W/m²

Heat flux density to surface outside skeletal structure 414,780.47 W/m²

Total heat flux density to surface $4.76\text{e}+06 \text{ W/m}^2$

Binder sublimation heat flux density $4.76\text{e}+06 \text{ W/m}^2$

Surface heat balance error 0.0011 W/m^2

At pressure 4.06 MPa

Experimental burning rate 13.63 mm/s

Calculated burning rate 13.67 mm/s

First Conditional Fuel

Surface temperature 878.66 K

Linear burning rate 65.92 mm/s

Mass decomposition rate $109.36 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Kinetic flame heat flux density to surface $1.44\text{e}+08 \text{ W/m}^2$

Kinetic flame height 1.96 μm

Average kinetic flame density 7.02 kg/m^3

Average kinetic flame temperature 2,292.83 K

Binder sublimation heat flux density $1.44\text{e}+08 \text{ W/m}^2$

Surface heat balance error 0.034 W/m^2

Secondary Conditional Fuel

Surface temperature 801.43 K

Linear burning rate 5.79 mm/s

Mass decomposition rate $9.61 \text{ kg}\cdot\text{s}^{-1}\cdot\text{m}^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 954,341.61 W/m²

Kinetic flame height 163.84 μm

Average kinetic flame density 10.17 kg/m^3

Average kinetic flame temperature 1,583.22 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 856,666.68 W/m²

Kinetic flame height 92.17 μm

Average kinetic flame density 13.46 kg/m^3

Average kinetic flame temperature 1,196.22 K

Heat flux density due to metal combustion, $2.39\text{e}+06 \text{ W/m}^2$

Effective thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity $0.4 \text{ W/(m}\cdot\text{K)}$

Radiative thermal conductivity $3.31\text{e}-06 \text{ W/(m}\cdot\text{K)}$

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness $8.43\text{e-}05$ m
Pore diameter $2.06\text{e-}09$ m
Diffusion flame heat flux density $357,223.57$ W/m²
Diffusion flame height 801.9 μm
Heat flux density to surface within skeletal structure $608,527.74$ W/m²
Heat flux density to surface outside skeletal structure $775,346.1$ W/m²
Total heat flux density to surface $6.25\text{e+}06$ W/m²
Binder sublimation heat flux density $6.25\text{e+}06$ W/m²
Surface heat balance error 0.0014 W/m²

At pressure 6.5 MPa

Experimental burning rate 18 mm/s
Calculated burning rate 17.94 mm/s

First Conditional Fuel

Surface temperature 900 K
Linear burning rate 119.93 mm/s
Mass decomposition rate 198.96 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface $2.69\text{e+}08$ W/m²
Kinetic flame height 1.04 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature $2,303.5$ K
Binder sublimation heat flux density $2.69\text{e+}08$ W/m²
Surface heat balance error -0.058 W/m²

Secondary Conditional Fuel

Surface temperature 809.08 K
Linear burning rate 7.52 mm/s
Mass decomposition rate 12.48 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface $1.14\text{e+}06$ W/m²
Kinetic flame height 136.58 μm
Average kinetic flame density 16.26 kg/m³
Average kinetic flame temperature $1,587.04$ K

Skeleton Structure Region

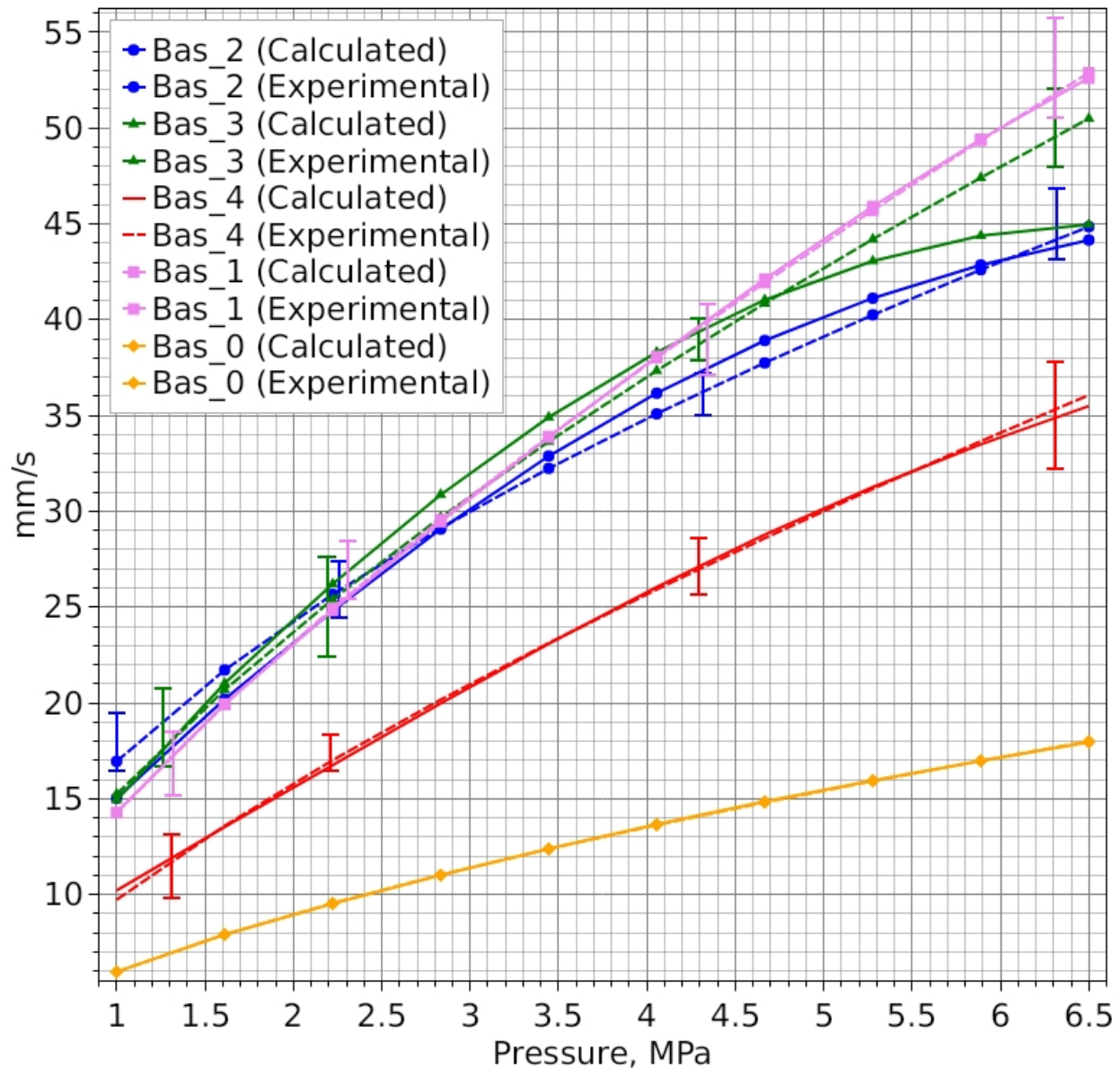
Kinetic flame heat flux density to surface $663,650.52$ W/m²
Kinetic flame height 117.82 μm
Average kinetic flame density 21.5 kg/m³
Average kinetic flame temperature $1,200.04$ K
Heat flux density due to metal combustion, $2.76\text{e+}06$ W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity $2.82\text{e-}06$ W/(m·K)
Conductive thermal conductivity balance error $-4.092294297874943\text{E-}08$
Skeleton layer thickness $7.18\text{e-}05$ m
Pore diameter $1.75\text{e-}09$ m
Diffusion flame heat flux density $276,946.06$ W/m²
Diffusion flame height $1,031.58$ μm
Heat flux density to surface within skeletal structure $431,724.14$ W/m²
Heat flux density to surface outside skeletal structure $995,594.53$ W/m²
Total heat flux density to surface $6.78\text{e+}06$ W/m²
Binder sublimation heat flux density $6.78\text{e+}06$ W/m²
Surface heat balance error -0.0012 W/m²

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	15.03 mm/s		14.93 mm/s		10.22 mm/s		14.3 mm/s		5.97 mm/s	
Difference	1.92 mm/s		0.31 mm/s		-0.49 mm/s		-0.034 mm/s		-3.87e-05 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.19 mm/s	8.02 mm/s	11.19 mm/s	11.19 mm/s	11.19 mm/s	3.53 mm/s	10.3 mm/s	7.71 mm/s	11.19 mm/s	2.69 mm/s
Surface Temperature	820.96 K	810.98 K	820.96 K	820.96 K	820.96 K	787.32 K	606.41 K	600 K	820.96 K	779.79 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.29e+07 W/m ²		2.29e+07 W/m ²		2.29e+07 W/m ²		1.56e+07 W/m ²		2.29e+07 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	8.77e+06 W/m ²		6.21e+07 W/m ²		4.65e+06 W/m ²		6.22e+06 W/m ²		1.81e+06 W/m ²	
Kinetic Flame Height (Skeleton)	8.89 μm		2.9 μm		9.75 μm		15.94 μm		44.77 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	188,989.63 W/m ²		135,882.41 W/m ²		7,226.83 W/m ²		181,739.18 W/m ²		559,449.24 W/m ²	
Kinetic Flame Height (Out-Skeleton)	822.28 μm		1,136.31 μm		3,524.15 μm		971.17 μm		283.35 μm	
Diffusion Flame Heat Flux Density	62,778.15 W/m ²		119,989.1 W/m ²		19,270.69 W/m ²		71,683.8 W/m ²		193,686 W/m ²	
Diffusion Flame Height	4,449.04 μm		2,377.75 μm		14,491.87 μm		4,277.12 μm		1,490.15 μm	
Metal Combustion Heat Flux Density	3.48e+06 W/m ²		3.7e+06 W/m ²		7.23e+06 W/m ²		2.09e+06 W/m ²		1.56e+06 W/m ²	

4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	36.17 mm/s		38.31 mm/s		26.05 mm/s		38.08 mm/s		13.67 mm/s	
Difference	-1.08 mm/s		-0.97 mm/s		-0.13 mm/s		-0.071 mm/s		-0.041 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	65.92 mm/s	16.34 mm/s	65.92 mm/s	20.12 mm/s	65.92 mm/s	7.89 mm/s	60.17 mm/s	17.49 mm/s	65.92 mm/s	5.79 mm/s
Surface Temperature	878.65 K	832.63 K	878.65 K	839.2 K	878.65 K	810.47 K	648.67 K	618.49 K	878.66 K	801.43 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.44e+08 W/m ²		1.44e+08 W/m ²		1.44e+08 W/m ²		9.72e+07 W/m ²		1.44e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.03e+07 W/m ²		8.14e+07 W/m ²		4.94e+06 W/m ²		1.95e+07 W/m ²		856,666.68 W/m ²	
Kinetic Flame Height (Skeleton)	7.4 μm		2.19 μm		8.71 μm		4.99 μm		92.17 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	340,975.74 W/m ²		277,268.43 W/m ²		11,524.81 W/m ²		295,548.88 W/m ²		954,341.61 W/m ²	
Kinetic Flame Height (Out-Skeleton)	449.41 μm		550.3 μm		2,009 μm		590.94 μm		163.84 μm	
Diffusion Flame Heat Flux Density	122,516.5 W/m ²		254,773.41 W/m ²		48,669.76 W/m ²		125,870.05 W/m ²		357,223.57 W/m ²	
Diffusion Flame Height	2,262.04 μm		1,112.68 μm		5,690.46 μm		2,421.15 μm		801.9 μm	
Metal Combustion Heat Flux Density	5.01e+06 W/m ²		4.45e+06 W/m ²		1.12e+07 W/m ²		3.37e+06 W/m ²		2.39e+06 W/m ²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	44.15 mm/s		44.96 mm/s		35.49 mm/s		52.6 mm/s		17.94 mm/s	
Difference	0.69 mm/s		5.54 mm/s		0.58 mm/s		0.28 mm/s		0.059 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	119.93 mm/s	19.27 mm/s	119.93 mm/s	21.83 mm/s	119.93 mm/s	10.49 mm/s	109.14 mm/s	23.46 mm/s	119.93 mm/s	7.52 mm/s
Surface Temperature	899.99 K	837.83 K	899.99 K	841.8 K	899.99 K	819.02 K	664.29 K	625.41 K	900 K	809.08 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.69e+08 W/m ²		2.69e+08 W/m ²		2.69e+08 W/m ²		1.81e+08 W/m ²		2.69e+08 W/m ²	
Kinetic Flame Heat Flux Density (Skeleton)	1.16e+07 W/m ²		1e+08 W/m ²		4.97e+06 W/m ²		2.82e+07 W/m ²		663,650.52 W/m ²	
Kinetic Flame Height (Skeleton)	6.48 μm		1.78 μm		8.49 μm		3.43 μm		117.82 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	447,892.88 W/m ²		395,670.75 W/m ²		13,240.17 W/m ²		342,065.01 W/m ²		1.14e+06 W/m ²	
Kinetic Flame Height (Out-Skeleton)	340.97 μm		384.97 μm		1,684.1 μm		508.56 μm		136.58 μm	
Diffusion Flame Heat Flux Density	104,686.86 W/m ²		236,700.4 W/m ²		37,003.36 W/m ²		94,537.47 W/m ²		276,946.06 W/m ²	
Diffusion Flame Height	2,642.33 μm		1,196.53 μm		7,461.42 μm		3,216.28 μm		1,031.58 μm	
Metal Combustion Heat Flux Density	5.46e+06 W/m ²		4.58e+06 W/m ²		1.31e+07 W/m ²		3.99e+06 W/m ²		2.76e+06 W/m ²	

Burning Rates - Group Optimization (All Propellants)



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.285341E-05$; $v = 0.52$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: $U = A \cdot p^v$, $A = 2.203002E-06$; $v = 0.64$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: $U = A \cdot p^v$, $A = 6.137894E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 9.000448E-07$; $v = 0.7$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: $U = A \cdot p^v$, $A = 1.720582E-06$; $v = 0.59$.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

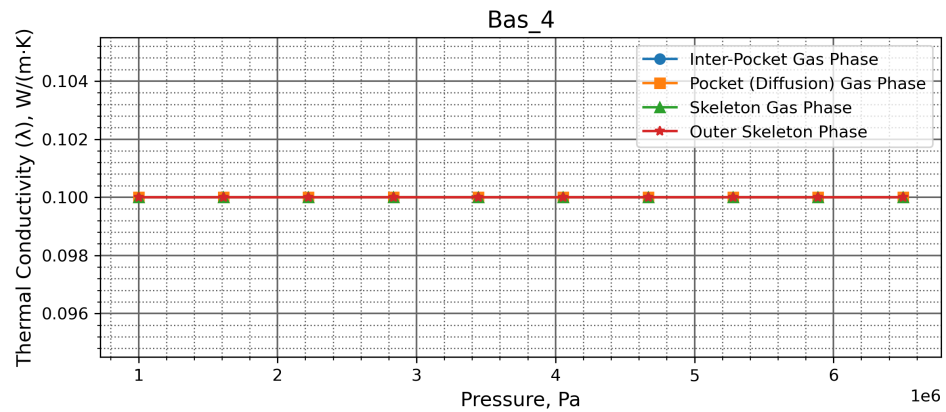
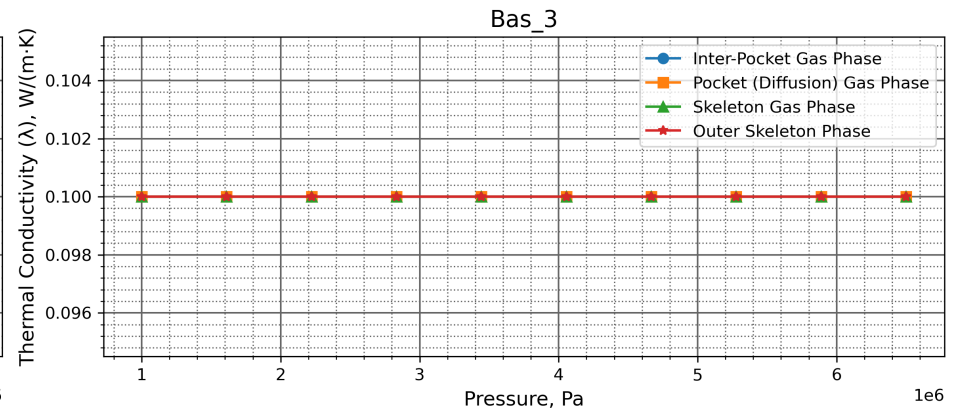
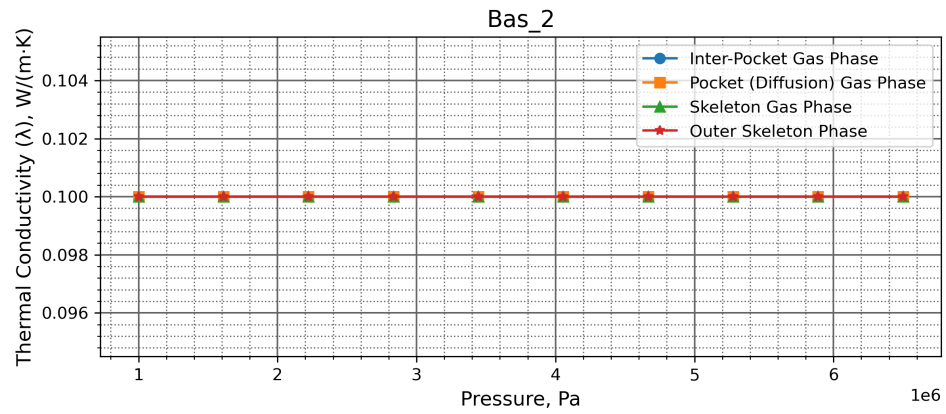
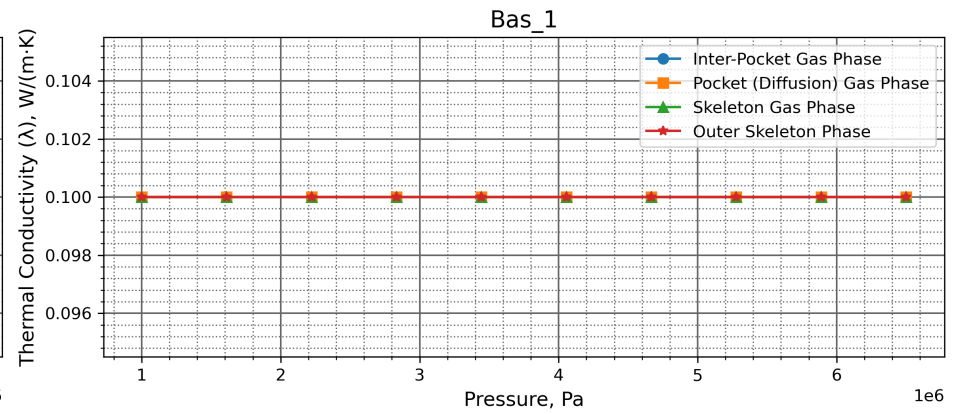
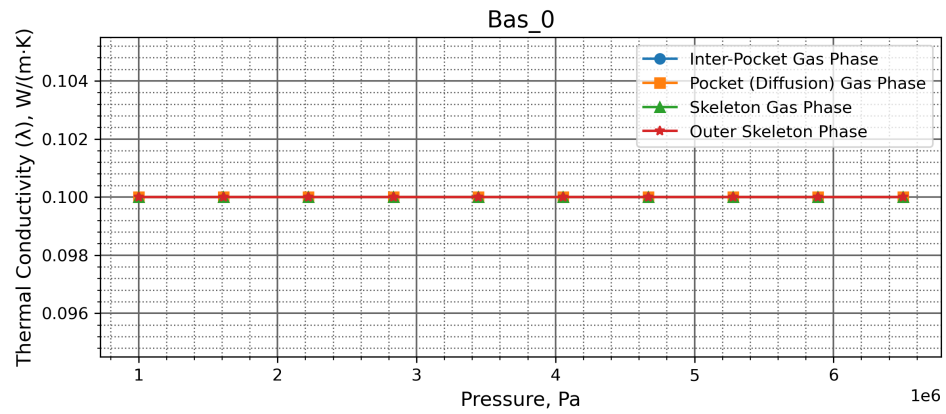
Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

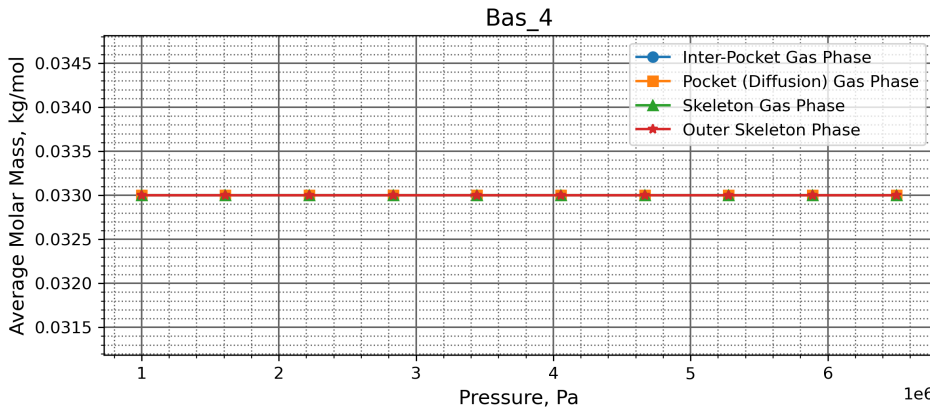
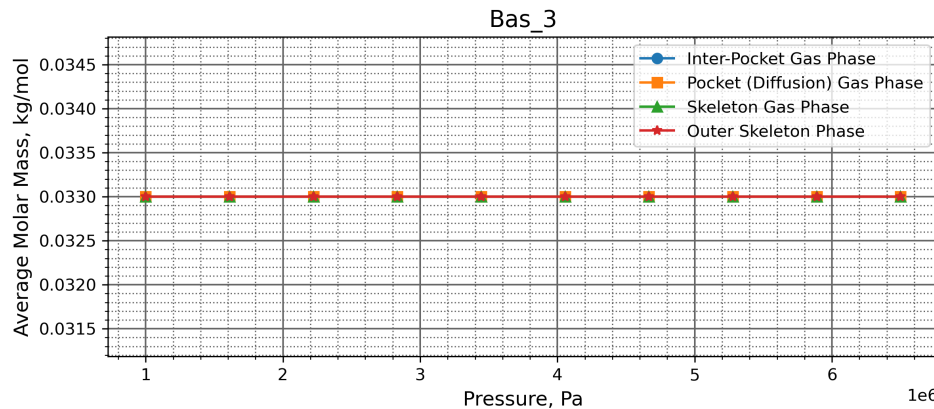
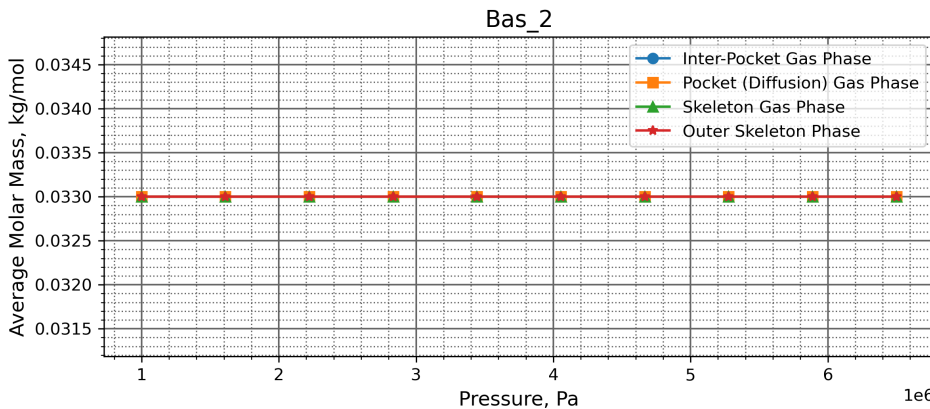
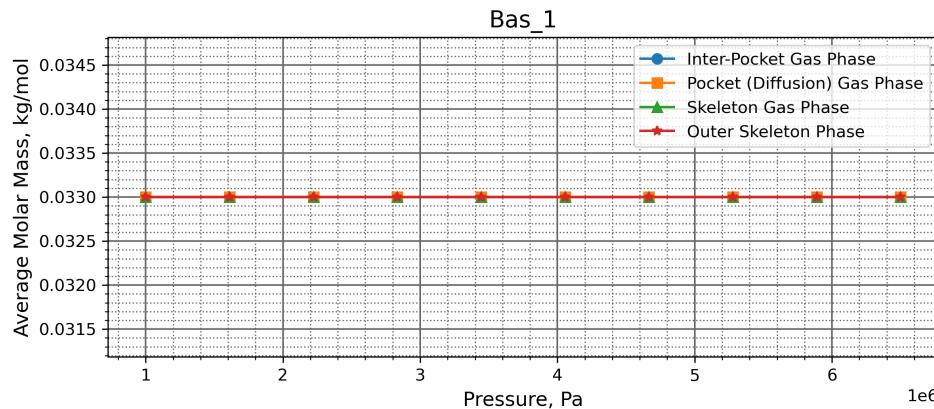
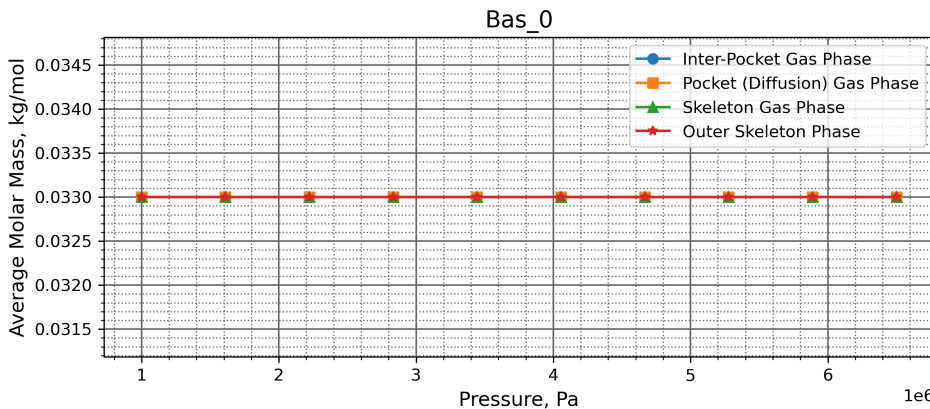
Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 2,365 K

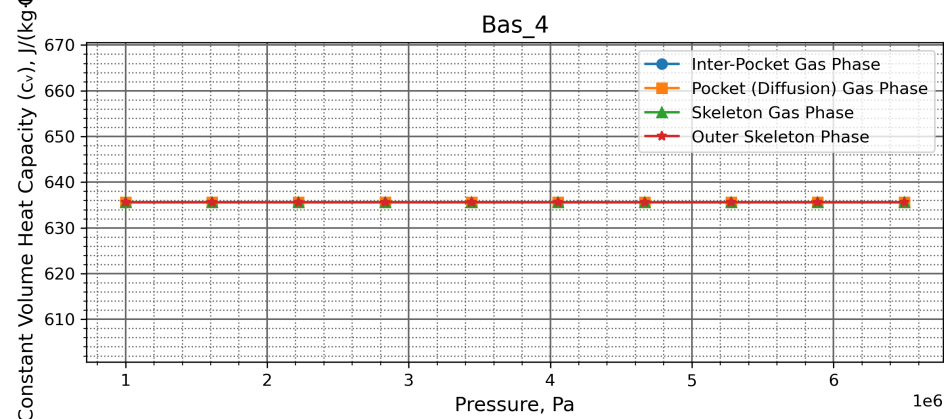
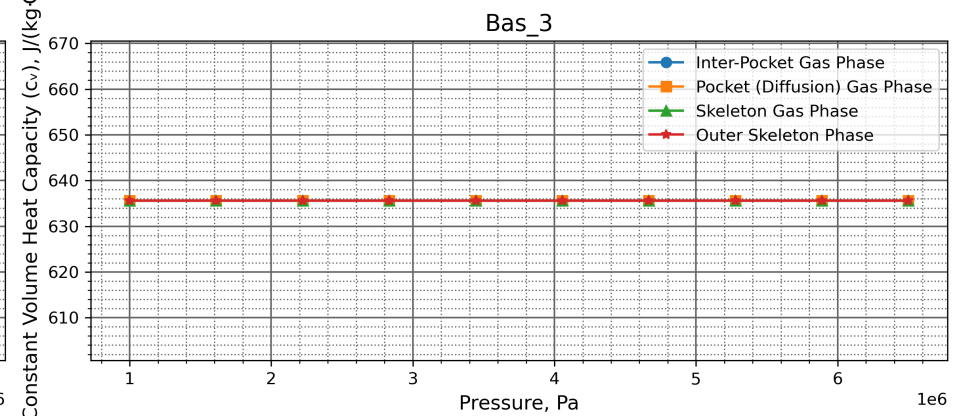
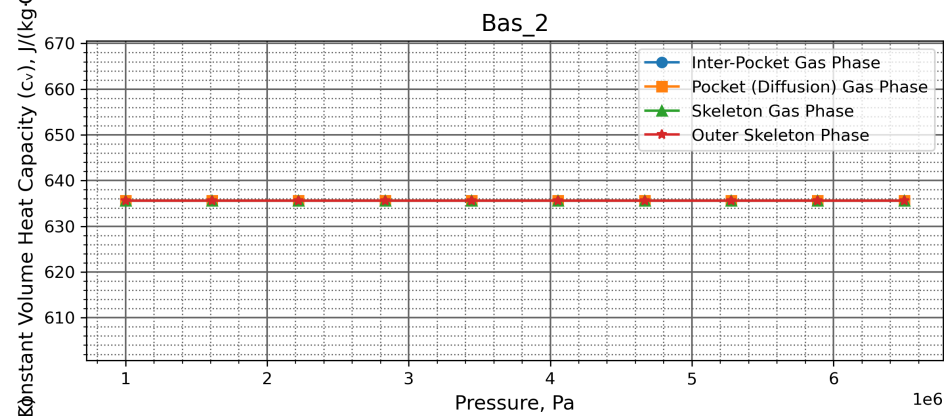
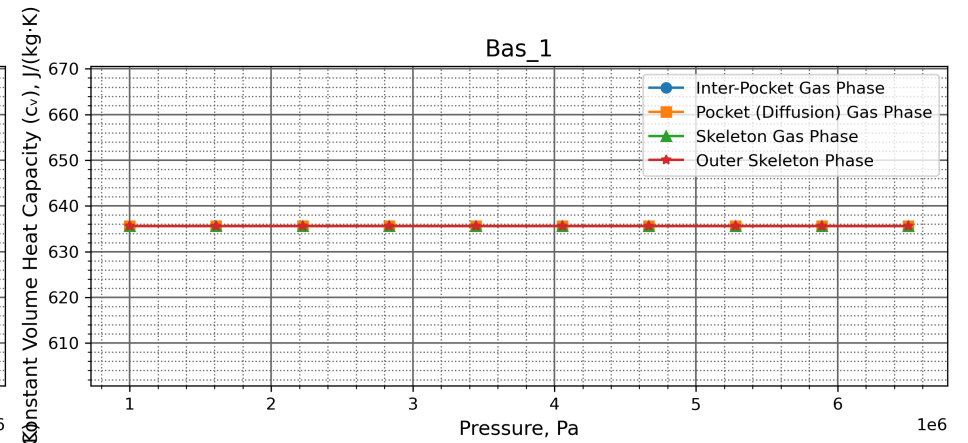
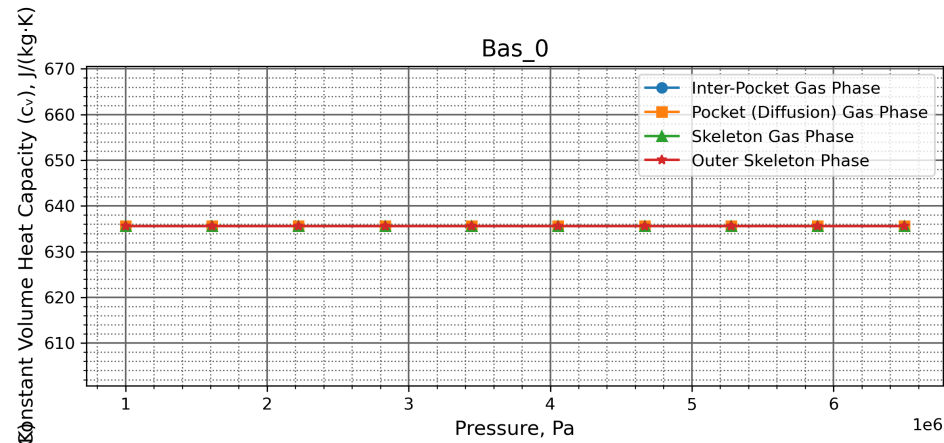
Thermal Conductivity (λ), W/(m·K)



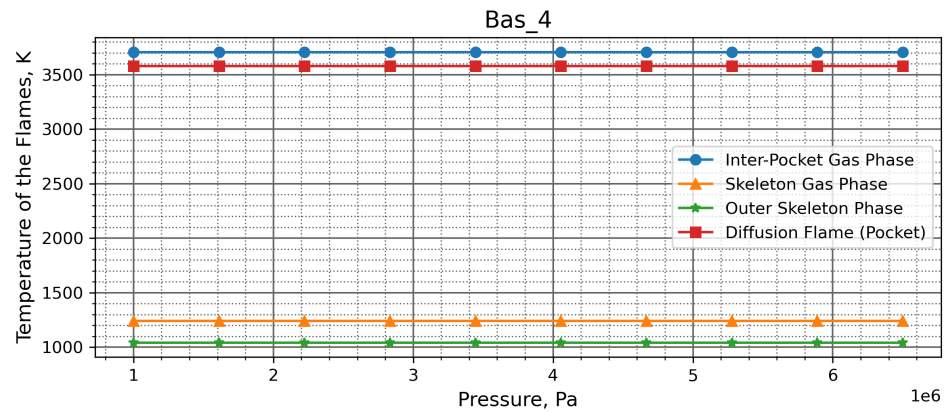
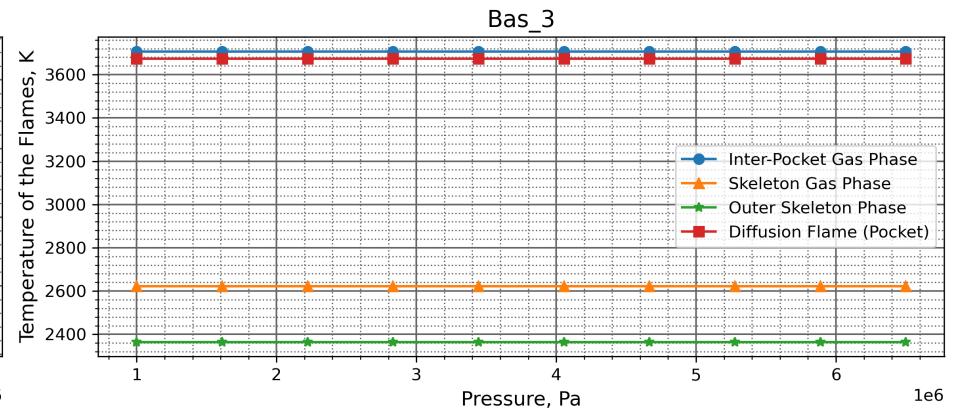
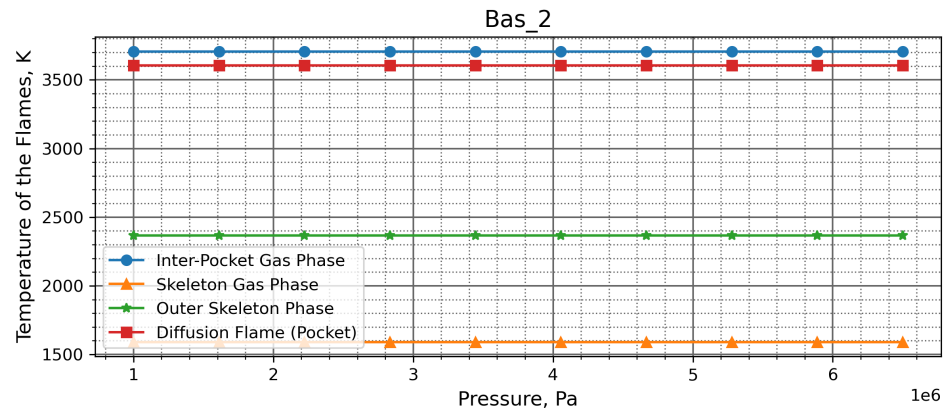
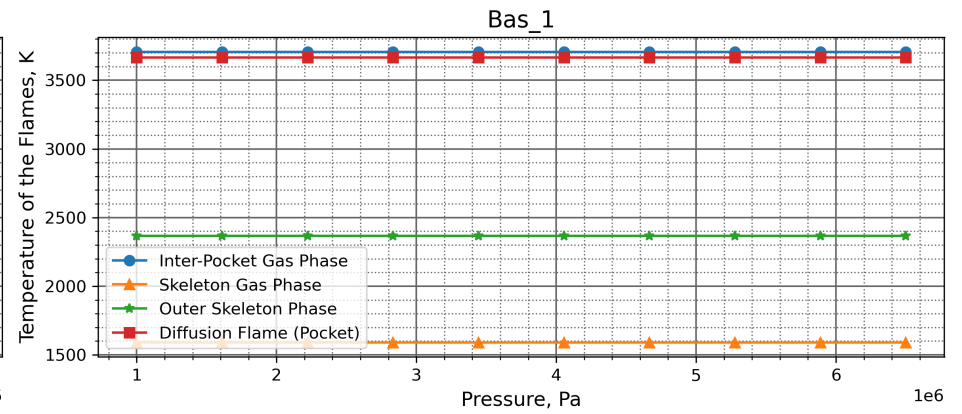
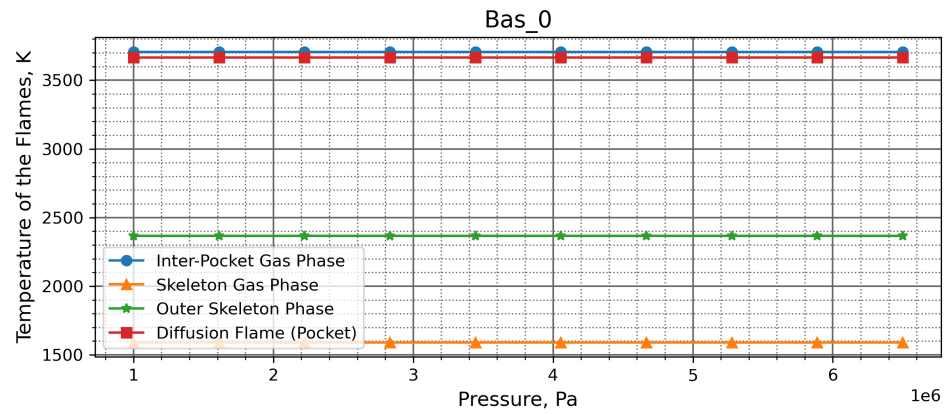
Average Molar Mass, kg/mol



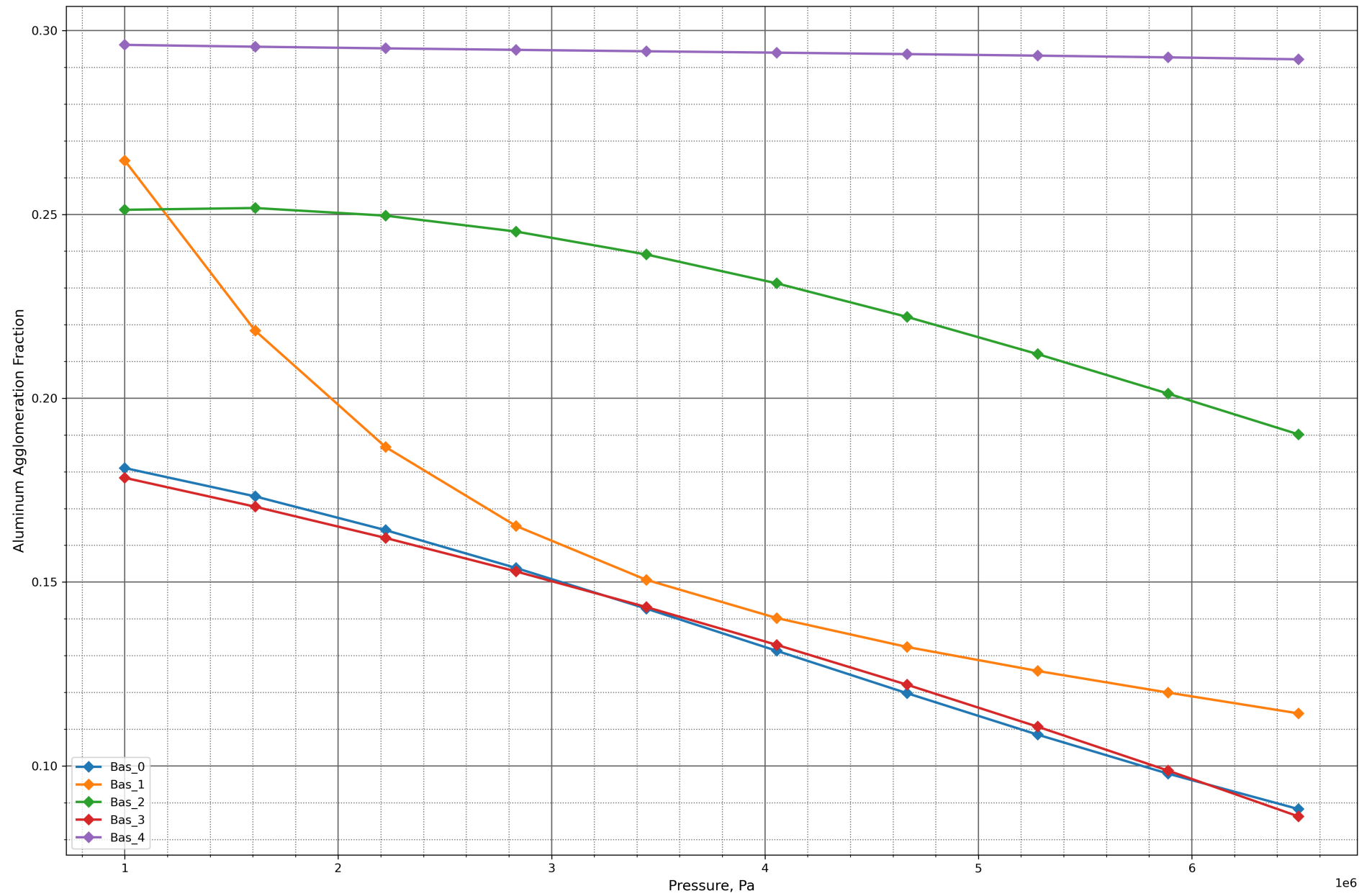
Constant Volume Heat Capacity (c_v), J/(kg·K)



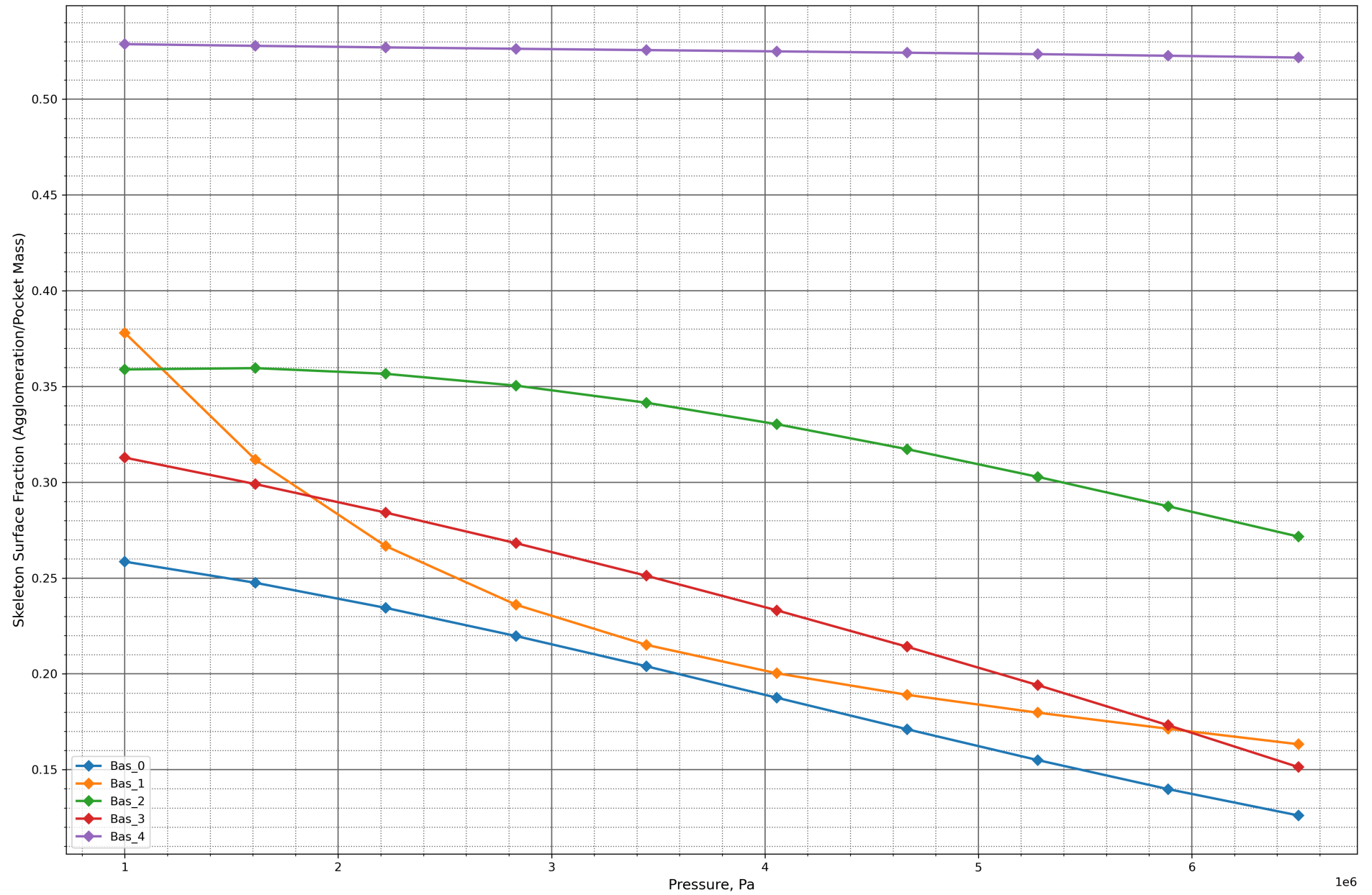
Temperature of the Flames, K



Aluminum Agglomeration Fraction



Skeleton Surface Fraction (Agglomeration/Pocket Mass)



Performance Report

System Information

Operating System: Unix 6.8.0.124 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.9 (.NET 10.0.9)
Architecture: X64 / X64

Total Execution Time

Total execution time: 09:01:39.349

Garbage Collection Information

Total Memory: 8473360 bytes
GC Pause Time Percentage: 0.10%
Generation 0 Collections: 30314
Generation 1 Collections: 2675
Generation 2 Collections: 298

Detailed Performance Metrics

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 18026519
Maximum execution time: 9.0343 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.8561226802802197 ms
Execution time standard deviation: 0.30267602851464187 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 15744889
Maximum execution time: 12.5082 ms
Minimum execution time: 0.0001 ms
Mean execution time: 0.3077868883038759 ms
Execution time standard deviation: 0.12212867731367684 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance

Call count: 13920923
Maximum execution time: 6.4546 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2909329798678158 ms
Execution time standard deviation: 0.13444166800669985 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 11.6929 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8482452991379782 ms
Execution time standard deviation: 0.30347476927221484 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 15511113
Maximum execution time: 6.4523 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.30453689252346855 ms
Execution time standard deviation: 0.12105951874617962 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance

Call count: 13664249
Maximum execution time: 9.5951 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.29285118150287537 ms
Execution time standard deviation: 0.13317841317257514 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 9.2032 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8483062068564456 ms
Execution time standard deviation: 0.30071943572020354 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 15573255
Maximum execution time: 6.4127 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3061070313624616 ms
Execution time standard deviation: 0.1210774102638504 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 13739642
Maximum execution time: 6.4355 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28963364183000356 ms
Execution time standard deviation: 0.13463152268858034 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 7.1636 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8580339594088114 ms
Execution time standard deviation: 0.30189006460996315 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 15569079
Maximum execution time: 6.6152 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3080183224967833 ms
Execution time standard deviation: 0.12108812049204996 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 13688729
Maximum execution time: 6.982 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29397691351041433 ms
Execution time standard deviation: 0.13382138902362495 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 9.0948 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8324698448007742 ms
Execution time standard deviation: 0.29937840491758055 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 15528460
Maximum execution time: 7.447 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29733225504655014 ms
Execution time standard deviation: 0.11892291613513986 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance

Call count: 13694313

Maximum execution time: 6.7471 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2828168650227576 ms

Execution time standard deviation: 0.13123514470276212 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 17926063

Maximum execution time: 12.8637 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8476541677667594 ms

Execution time standard deviation: 0.30049676072328824 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 15571326

Maximum execution time: 11.5266 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.30380025941914507 ms

Execution time standard deviation: 0.12057697733352486 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance

Call count: 13738611

Maximum execution time: 6.4136 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2908169487657311 ms

Execution time standard deviation: 0.13347432600248754 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 17926063

Maximum execution time: 12.1179 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8463749558898178 ms

Execution time standard deviation: 0.3022639171643341 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 15641094

Maximum execution time: 9.4396 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3020683956186193 ms

Execution time standard deviation: 0.1210183212888422 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance

Call count: 13825434

Maximum execution time: 5.434 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2908492573397381 ms

Execution time standard deviation: 0.1318923567411467 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 17926063

Maximum execution time: 16.7727 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8054726758628672 ms

Execution time standard deviation: 0.29231731824912816 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 15481728

Maximum execution time: 11.7073 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28941089411985566 ms
Execution time standard deviation: 0.11667953534829424 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance

Call count: 13621559
Maximum execution time: 8.3036 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2759924260651406 ms
Execution time standard deviation: 0.1284810544333969 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 12.8285 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8617820604892534 ms
Execution time standard deviation: 0.303405942839588 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 15568877
Maximum execution time: 12.3414 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3087057776228753 ms
Execution time standard deviation: 0.12165997986483372 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance

Call count: 13689276
Maximum execution time: 9.4205 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29687369089495297 ms
Execution time standard deviation: 0.13377126348017251 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 10.2602 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8550964915608916 ms
Execution time standard deviation: 0.3032646534637226 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 15522263
Maximum execution time: 4.7462 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3061659775253978 ms
Execution time standard deviation: 0.12132863476065961 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance

Call count: 13667684
Maximum execution time: 6.3944 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.294318771695297 ms
Execution time standard deviation: 0.1344218986976339 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 13.4051 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8748980016360317 ms

Execution time standard deviation: 0.30644976660932477 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 15567668

Maximum execution time: 8.6066 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.31380133738084043 ms

Execution time standard deviation: 0.12260773577505808 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance

Call count: 13688429

Maximum execution time: 6.4035 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.29936336529924307 ms

Execution time standard deviation: 0.1359954053730171 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 17926063

Maximum execution time: 8.0668 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8510409164411482 ms

Execution time standard deviation: 0.30600648968073096 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 15465308

Maximum execution time: 6.4455 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3079180161299972 ms

Execution time standard deviation: 0.12222387505450374 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance

Call count: 13607372

Maximum execution time: 6.398 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2930836479887579 ms

Execution time standard deviation: 0.13435282539830526 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 17926063

Maximum execution time: 11.1184 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.8677152534496383 ms

Execution time standard deviation: 0.30147717885457376 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 15618343

Maximum execution time: 9.096 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.3123162639148262 ms

Execution time standard deviation: 0.12116052710164298 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance

Call count: 13821261

Maximum execution time: 4.3228 ms

Minimum execution time: 0.0002 ms

Mean execution time: 0.2968664413471069 ms

Execution time standard deviation: 0.13290674310213316 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 7.1989 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.872529873882465 ms
Execution time standard deviation: 0.3100476937238763 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 15531815
Maximum execution time: 7.2061 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.31376189380957165 ms
Execution time standard deviation: 0.12329758300952487 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 13702481
Maximum execution time: 8.5302 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29811971342995464 ms
Execution time standard deviation: 0.13656266883450735 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 10.4878 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8454888036876576 ms
Execution time standard deviation: 0.3048189815057205 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 15532242
Maximum execution time: 7.4054 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.30367556017994973 ms
Execution time standard deviation: 0.12098769181390805 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance

Call count: 13680047
Maximum execution time: 8.4475 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28969860372553996 ms
Execution time standard deviation: 0.13311013450560222 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 17926063
Maximum execution time: 8.6583 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8411204786739942 ms
Execution time standard deviation: 0.3053838896964793 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 15396863
Maximum execution time: 7.0469 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3029901898003584 ms
Execution time standard deviation: 0.12063643734725829 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance

Call count: 13493729
Maximum execution time: 6.4395 ms
Minimum execution time: 0.0002 ms

Mean execution time: 0.2870544533908578 ms
Execution time standard deviation: 0.13326021229353283 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 12.8049 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8435435960255729 ms
Execution time standard deviation: 0.3045333334587889 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 15462563
Maximum execution time: 7.0876 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3027973809452116 ms
Execution time standard deviation: 0.12083508345769242 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 13537979
Maximum execution time: 9.2251 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29018988270696133 ms
Execution time standard deviation: 0.1336845870089915 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 7.1805 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8544258258658434 ms
Execution time standard deviation: 0.30625768276957865 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 15455567
Maximum execution time: 5.4291 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3066778977762225 ms
Execution time standard deviation: 0.12142994318318835 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 13608710
Maximum execution time: 6.3286 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2920605889463171 ms
Execution time standard deviation: 0.1344010284282241 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 10.1728 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8318100851648719 ms
Execution time standard deviation: 0.2990375463453897 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 15508063
Maximum execution time: 7.5152 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2990309873644653 ms
Execution time standard deviation: 0.11869425813421815 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 13654298
Maximum execution time: 6.8577 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.28386983320565223 ms
Execution time standard deviation: 0.13134076453763024 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 9.505 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8607504618385867 ms
Execution time standard deviation: 0.3026922048901286 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 15560185
Maximum execution time: 31.1563 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3086406638546066 ms
Execution time standard deviation: 0.12191567504998721 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 13664023
Maximum execution time: 19.4227 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29157267489961797 ms
Execution time standard deviation: 0.1355526919370538 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 8.6255 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.850673920999669 ms
Execution time standard deviation: 0.30108205925899434 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 15615541
Maximum execution time: 4.6553 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.3051535873717417 ms
Execution time standard deviation: 0.12092458114051706 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 13801470
Maximum execution time: 4.3971 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29363870962297023 ms
Execution time standard deviation: 0.1318589062374215 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 13.6051 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8712536405734972 ms
Execution time standard deviation: 0.3054359495892275 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 15535098
Maximum execution time: 6.9918 ms

Minimum execution time: 0.0002 ms
Mean execution time: 0.3137419035657546 ms
Execution time standard deviation: 0.1222832268579139 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 13700244
Maximum execution time: 7.3032 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29879038739017316 ms
Execution time standard deviation: 0.1347719139330656 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 10.2414 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8343196975877702 ms
Execution time standard deviation: 0.3001864756953385 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 15543338
Maximum execution time: 8.6188 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.29977221713254243 ms
Execution time standard deviation: 0.12023473423376597 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 13678828
Maximum execution time: 6.608 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2872183466888228 ms
Execution time standard deviation: 0.1310866649762081 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 17926063
Maximum execution time: 18.2685 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.8564108118441508 ms
Execution time standard deviation: 0.3026467503214945 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 15534926
Maximum execution time: 10.4826 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.30711960155457124 ms
Execution time standard deviation: 0.1209153463248063 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 13682651
Maximum execution time: 6.0942 ms
Minimum execution time: 0.0002 ms
Mean execution time: 0.2933042756772419 ms
Execution time standard deviation: 0.13398399747610537 ms